

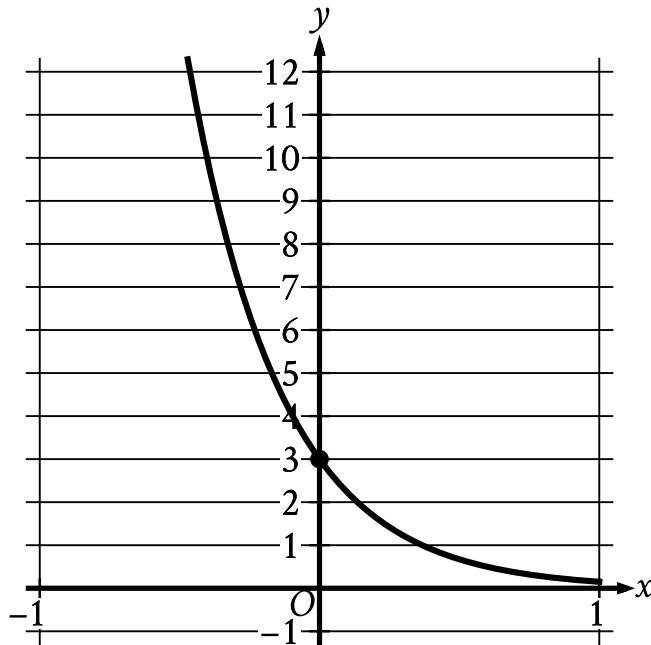
● EASY

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

01



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends down sharply to the point (0 comma 3).
 - In quadrant 1, the curve trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
 - (0 comma 3)
 - approximately (0.5 comma 0.7)

The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is $(0, y)$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$hx = x^2 - 3$$

Which table gives three values of x and their corresponding values of hx for the given function h ?

A.

x	1	2	3
hx	4	5	6

B.

x	1	2	3
hx	-2	1	6

C.

x	1	2	3
hx	-1	1	3

D.

x	1	2	3
hx	-2	1	3

The kinetic energy, in joules, of an object with mass 9 kilograms traveling at a speed of v meters per second is given by the function K , where $Kv = \frac{9}{2}v^2$. Which of the following is the best interpretation of $K34 = 5,202$ in this context?

- A. The object traveling at 34 meters per second has a kinetic energy of 5,202 joules.
- B. The object traveling at 340 meters per second has a kinetic energy of 5,202 joules.
- C. The object traveling at 5,202 meters per second has a kinetic energy of 34 joules.
- D. The object traveling at 23,409 meters per second has a kinetic energy of 34 joules.

$$P_t = 24.81.036^t$$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement “ P_8 is approximately equal to 32.91”?

- A. In 1984, the predicted population of this country was approximately 8 million.
- B. In 1984, the predicted population of this country was approximately 32.91 million.
- C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.
- D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.

Q5

GRID-IN ADVANCED MATH

The function f is defined by $f(x) = 6 + \sqrt{x}$. What is the value of $f(36)$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = -\frac{1}{4}x^2 + 2x + 29$$

The given equation models a company's scheduled deliveries over 8 months, where y is the estimated number of scheduled deliveries x months after the end of May 2012, where $0 \leq x \leq 8$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

- A. At the end of May 2012, the estimated number of scheduled deliveries was 0.
- B. At the end of May 2012, the estimated number of scheduled deliveries was 29.
- C. At the end of June 2012, the estimated number of scheduled deliveries was 0.
- D. At the end of June 2012, the estimated number of scheduled deliveries was 29.

Q7

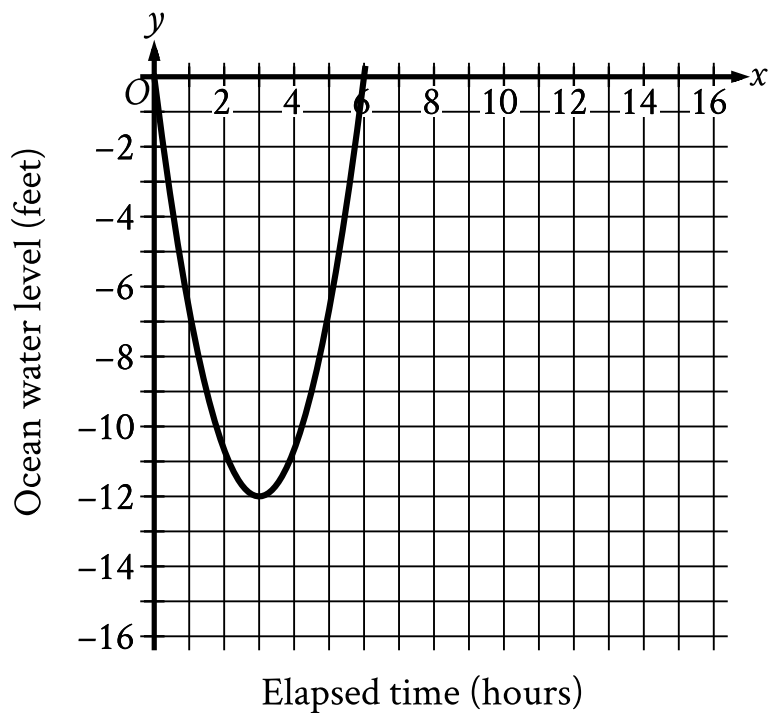
GRID-IN ADVANCED MATH

The y -intercept of the graph of $y = x^2 + 31$ in the xy -plane is $0, y$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The parabola opens upward.
- The parabola passes through the following points:
 - (0 comma 0)
 - (3 comma negative 12)
 - (6 comma 0)

Scientists recorded data about the ocean water levels at a certain location over a period of 6 hours. The graph shown models the data, where $y = 0$ represents sea level. Which table gives values of x and their corresponding values of y based on the model?

A.

x	y
0	-12
0	3
3	6

B.

x	y
0	0
3	12
0	-6

C.

x	y
0	0
3	-12
6	0

D.

x	y
0	0
12	3
-6	0

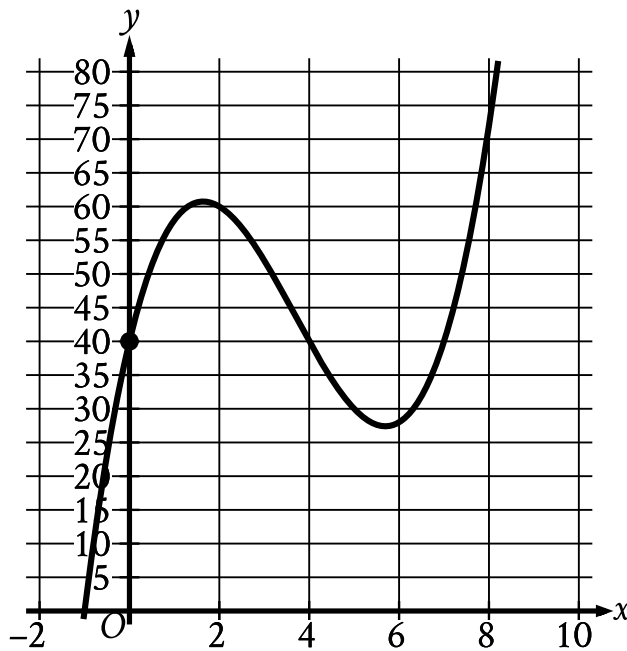
The function f is defined by $f(x) = x^3 + 9$. What is the value of $f(2)$?

A. 14

B. 15

C. 17

D. 18



- Moving from left to right, the curve passes from quadrant 2 to quadrant 1.
- The curve has 1 relative maximum and 1 relative minimum.
- The curve passes through the following points:
 - (0 comma 40)
 - (4 comma 40)
 - (7 comma 40)

The y -intercept of the graph shown is x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.